Week 5: Support Ticket Router

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Outcome

Fine-tuning improved held-out routing accuracy from 35.90% (42/117) to 84.62% (99/117), a gain of 48.72 percentage points. Macro F1 increased from 0.3004 to 0.8272. The final model generated valid labels for all 117 validation tickets. Training, adapter merging, evaluation and verified checkpoint preservation completed successfully.

Business purpose and data

The router assigns IT helpdesk tickets to seven queues, supporting faster initial triage. This project evaluates a prototype; it does not integrate with a production helpdesk or establish production cost, latency, or reliability.

The official dataset contains 585 tickets. The audit found no empty text, unsupported labels, normalized duplicates or conflicting labels. A stratified 80/20 split with seed 42 produced 468 training tickets and 117 validation tickets, with zero normalized-text or row-ID overlap. Only training rows entered the fine-tuning dataset.

Training / validation counts: Support general 120 / 30; Fileservice 110 / 28; O365 74 / 18; EOL 46 / 12; Software 46 / 12; Active Directory 39 / 9; Computer-Services 33 / 8.

Training and compatibility correction

Model: Qwen/Qwen3-1.7B-Base. Workflow: LLaMA Board in Colab on a Tesla T4, using the existing compute allowance. Final settings: LoRA rank 8, alpha 16, all linear targets, dropout 0; learning rate 0.0001; three epochs; batch size 1 with gradient accumulation 8; fp16; cosine schedule with warmup ratio 0.1; seed 42; cutoff 1024. The longest formatted training example contained 177 tokens.

The initial qwen3_nothink run completed but generated category prefixes followed by unwanted continuation, yielding 0/7 valid smoke outputs. Tokenizer vocabulary and prompt tokens matched training. The corrected run uses LLaMA-Factory's supported default template and retains the base model's native <|endoftext|> EOS (151643). This change was made using smoke diagnostics before any fine-tuned held-out evaluation; parsing was not relaxed.

An intermediate corrected run lost its temporary Colab runtime before final evaluation. The final run therefore wrote training checkpoints and live evidence directly to Drive. It completed all 177 optimization steps in 584.023 seconds. First/final logged loss was 7.8944 / 0.0959; overall reported training loss was 0.6160. Earlier diagnostic evidence is retained separately.

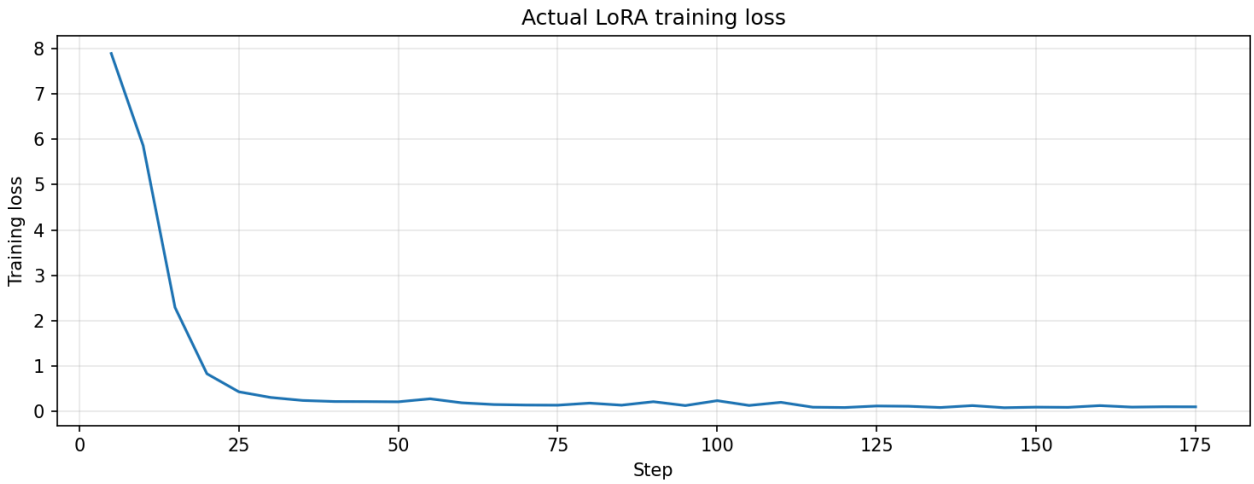


Figure 1. Actual loss measurements from the completed final run.

Held-out evaluation

Both approaches used the same 117 validation row IDs. The baseline selected one A–G option from first-token logits; the fine-tuned model generated an exact category label greedily. Consequently, the accuracy difference compares the complete routing approaches, including their decoding protocols, rather than isolating the effect of weights alone. Invalid outputs were counted as errors, and class order was explicit.

Baseline: accuracy 35.90%; macro F1 0.3004; invalid outputs 0.

Fine-tuned: accuracy 84.62%; macro F1 0.8272; invalid outputs 0.

Accuracy improvement: 48.72 percentage points, or 57 additional correct tickets.

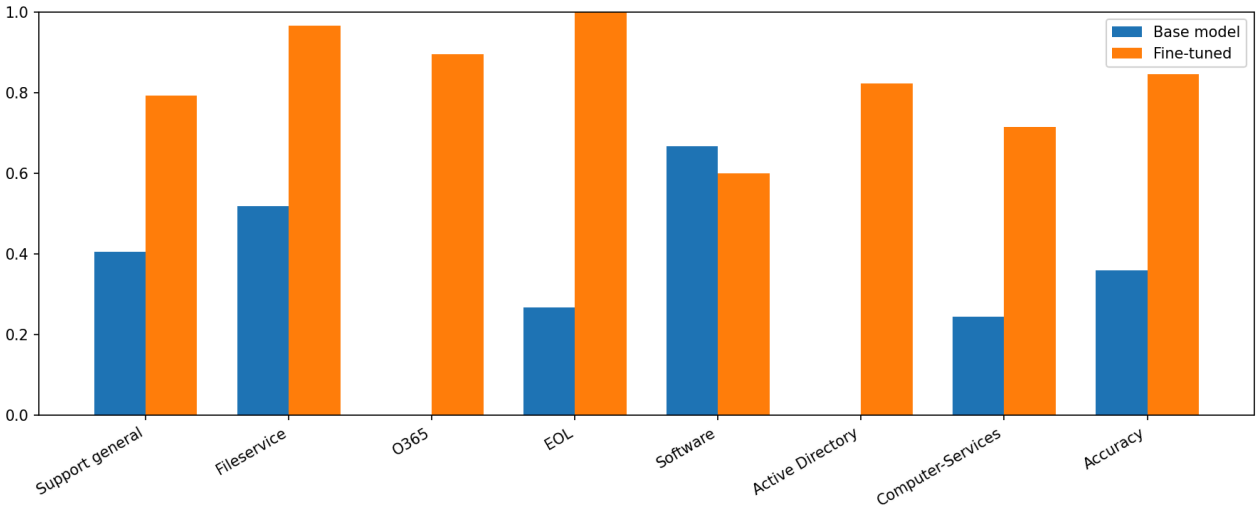


Figure 2. Per-class F1 and overall accuracy for the baseline and fine-tuned approaches.

Fine-tuned per-class F1 (validation support): Support general 0.7925 (30); Fileservice 0.9655 (28); O365 0.8947 (18); EOL 1.0000 (12); Software 0.6000 (12); Active Directory 0.8235 (9); Computer-Services 0.7143 (8). Full precision, recall and confusion counts are saved in metrics.json.

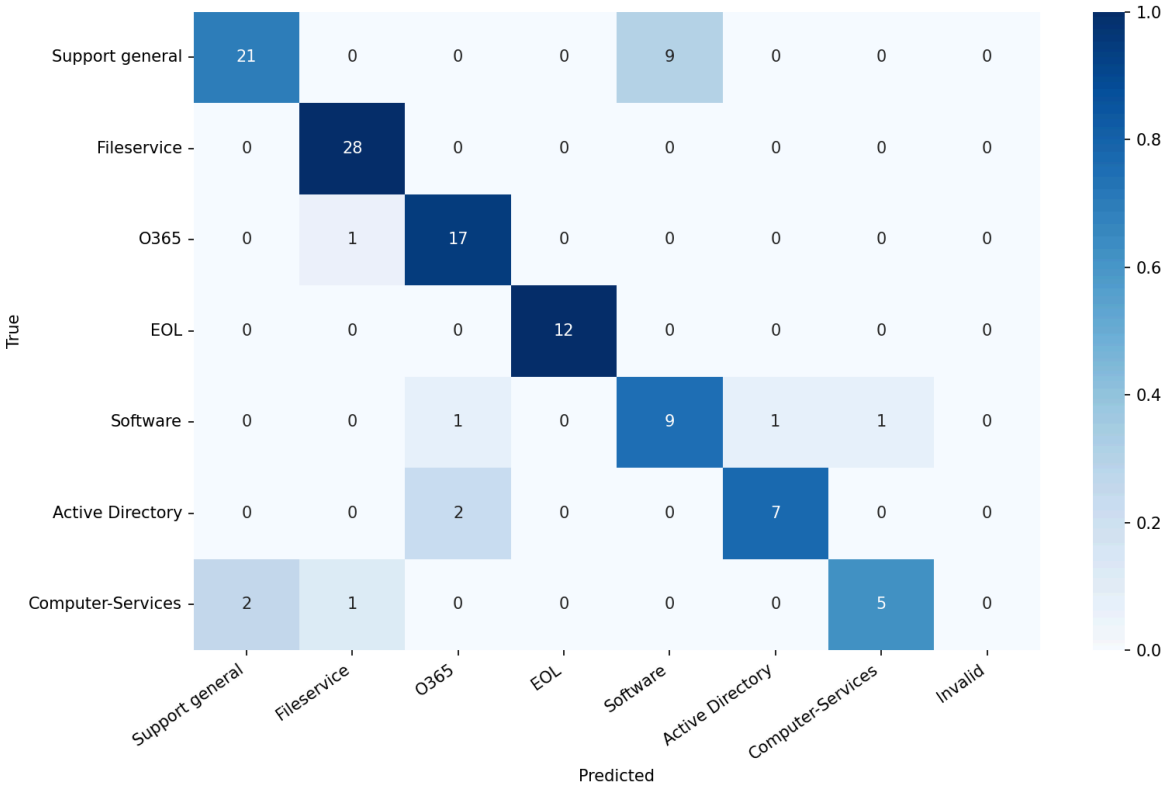


Figure 3. Cell annotations show ticket counts; shading shows the proportion within each true class.

Nine of the 18 errors were Support general tickets predicted as Software. Software precision was 50%, making this the clearest boundary to improve. Computer-Services recall was 62.5%. The small EOL sample achieved 12/12 correct, which does not establish perfect performance on future tickets.

Smoke tests, merge checks and review

All seven smoke outputs were valid labels, and six matched their expected category. The account-creation example was predicted as Support general instead of Active Directory. This error remains recorded; expected labels were not edited. Adapter-loaded and merged-model raw outputs matched on all seven smoke cases.

Fifteen local regression tests passed. Saved predictions independently reproduced both accuracy and macro-F1 values, every per-class result, and the confusion matrices. Independent review found no actionable issues in the new execution evidence or compatibility changes. It confirmed split integrity, training-payload fidelity, completed training and the recorded merge results.

The adapter, merged model, tokenizer and evidence were copied to Drive with file-size and SHA-256 verification. The merged model contains six files totaling 3,452,610,781 bytes; its model.safetensors file is 3,441,185,296 bytes. The independent reviewer checked manifest consistency and local evidence hashes, but did not separately download and rehash the multi-GB checkpoint.

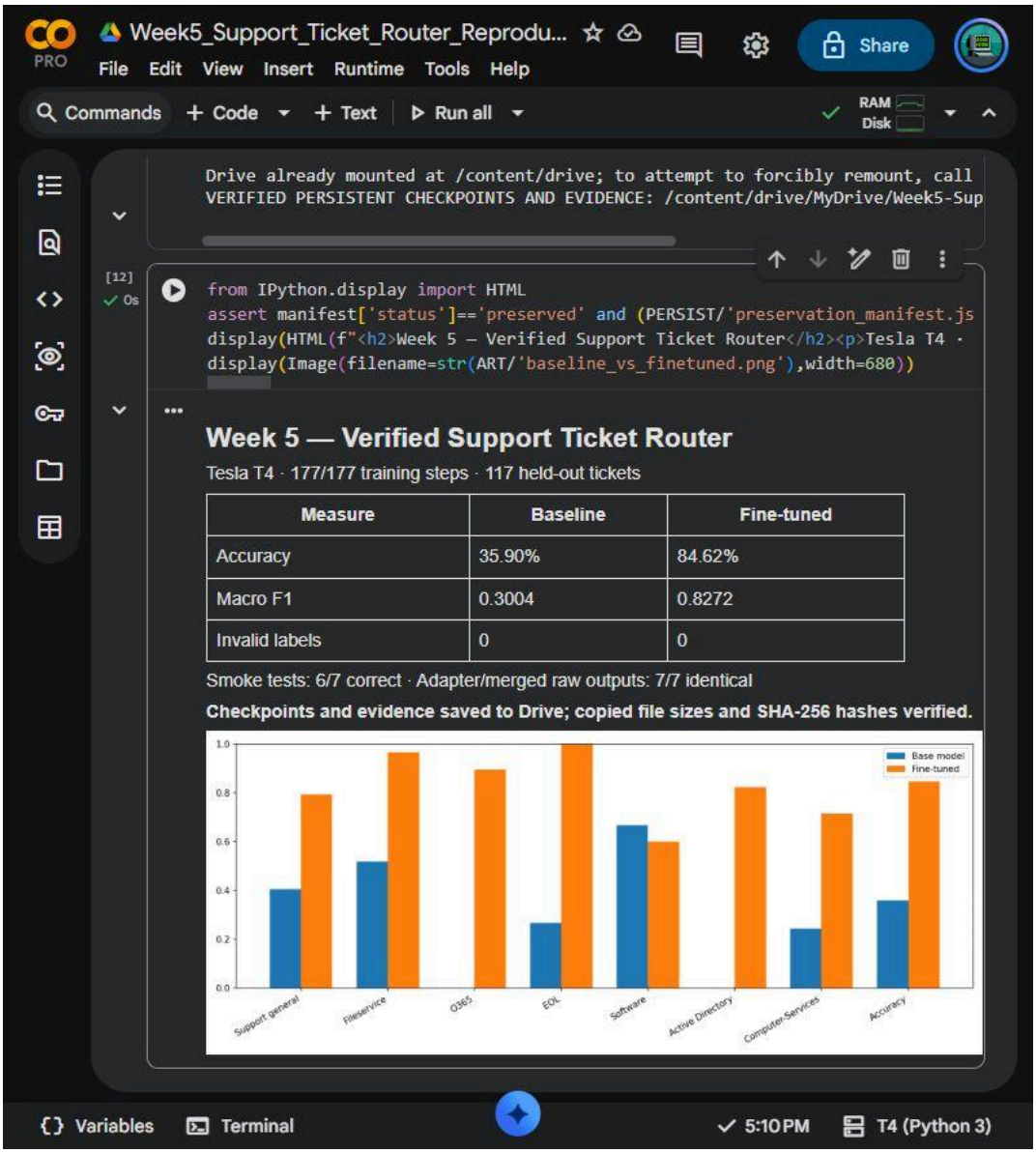


Figure 4. Authentic Colab output showing completed training, measured results and verified preservation.

Limitations and next steps

This is one seeded holdout, with only 8–30 validation tickets per class. It is not a separate production test or a multi-seed study. Labels overlap in wording: some application and account requests are assigned to broader queues. Review those labeling boundaries, expand the smallest classes, and evaluate a new untouched test set before deployment. Use human review for ambiguous routing; no calibrated confidence, latency benefit or operating-cost saving is claimed.

Reproduction and saved assets

Executed notebook: https://colab.research.google.com/drive/1hX7lF46sAjOaq-Jbhth56AHEQHYTFXLm

Checkpoints and evidence:
https://drive.google.com/drive/folders/1A-web_Oehc26cDI0lYCik93lsb-excPf

Week 5 project folder: https://drive.google.com/drive/folders/1eReAPODspet6nHwzFnZNI_LNwjAuRf3I

Earlier diagnostic notebook:
https://colab.research.google.com/drive/1AvJx4cNOX04fiUEPtxZOhEs9KTX7GXV_

For a clean rerun, use notebooks/Week5_Support_Ticket_Router.ipynb in the local repository, mount Drive, then follow the LLaMA Board settings printed by preflight. Choose a fresh output directory for a new experiment. After Start, the completion gate can wait for training while the evaluation and preservation cells are queued. The executed notebook preserves the actual final outputs.

Official assignment source:
https://github.com/The-Gen-Academy/5A-Fine-Tune-a-Support-Ticket-Router

LLaMA-Factory base-model template guidance: https://github.com/hiyouga/LlamaFactory

Source revision: 05f7ea9643495741bda4e272bd1977105c3ad529

Model revision: ea980cb0a6c2ae4b936e82123acc929f1cec04c1

LLaMA-Factory revision: f6355ab6a426eedb800742ecf61896f7bff95ebb

Dataset SHA-256: c55c98b462f1af9b2f5e4ad8a7460a6b598e55bcfa42da1b57dc9e921e82783f

Local source and evidence are retained in D:/Documents/Code Repos/The Gen Academy Mastering Agentic AI/Week5-Support-Ticket-Router. Large model weights are excluded from Git. No GitHub publication or course-form submission was performed.